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Mapping and Prioritizing Barriers to Sustainable Green Supply Chain Management: An Integrated Fuzzy MADM Approach in Manufacturing Enterprises

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Abstract

Sustainable Green Supply Chain Management (GSCM) is increasingly recognized as a critical strategy for achieving long-term competitive advantage and environmental performance in manufacturing industries. Despite its importance, various barriers hinder the effective implementation of GSCM. This study aims to identify, analyze, and prioritize the critical barriers that impede sustainable GSCM practices in manufacturing enterprises. To achieve this, an integrated Multi-Attribute Decision-Making (MADM) framework combining fuzzy Decision-Making Trial and Evaluation Laboratory (DEMATEL), fuzzy Evaluation based on Distance from Average Solution (EDAS), fuzzy COMplex PROportional ASsessment (COPRAS), and the Copeland method was employed. Initially, barriers were identified through comprehensive literature reviews and expert interviews. Subsequently,

fuzzy DEMATEL was applied to analyze the causal relationships among these barriers, and fuzzy EDAS and COPRAS methods were utilized to rank the barriers based on their relative significance.

Finally, the Copeland method aggregated the rankings to determine the most critical barriers. Results indicate that the lack of financial resources, inadequate regulatory frameworks, insufficient technological innovation, and weak organizational commitment are the most significant barriers to progress. The study offers practical insights for manufacturing managers and policymakers,

suggesting targeted interventions such as employee training programs, fostering collaborative relationships, and enhancing governmental support to overcome these barriers effectively.

Keywords: Green supply chain management, Barriers, Green human resource management, DEMATEL, EDAS, COPRAS, Manufacturing enterprises.

1|Introduction

The transition toward sustainability has become imperative for manufacturing enterprises worldwide [1–3]. Amid growing environmental concerns, resource depletion, and regulatory pressures, industries are

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increasingly urging the adoption of practices that minimize ecological impact while maintaining economic performance. Among these practices, Green Supply Chain Management (GSCM) has emerged as a strategic approach that integrates environmental thinking into every stage of the supply chain — from product design and material sourcing to production, distribution, and end-of-life management [4], [5]. Despite its proven benefits in reducing carbon footprints, enhancing resource efficiency, and strengthening stakeholder trust, the adoption of GSCM remains uneven, especially in emerging economies and industrial contexts where structural, behavioral, and operational challenges persist [6–10].

Despite its strategic relevance, the implementation of GSCM is often hindered by a multitude of interrelated barriers [11–13]. These obstacles are not limited to technological or financial limitations but also include organizational misalignment, lack of regulatory support, insufficient employee engagement, and underdeveloped green capabilities within firms [14–16]. In manufacturing contexts, particularly those operating in resource-constrained environments, the complexity of these barriers is further amplified by fragmented supply networks, outdated infrastructure, and inconsistent managerial commitments. While prior studies have explored some of these impediments individually, a holistic and prioritized understanding of the most critical barriers remains elusive [17–20]. This knowledge gap restricts the ability of managers and policymakers to allocate resources effectively and to design interventions tailored to context-specific challenges. Addressing the multifaceted barriers to GSCM requires structured data-driven approaches that can handle complexity, interdependency, and uncertainty. Conventional managerial tools often fail to provide meaningful insights when decision variables are interrelated and information is imprecise. To overcome such limitations, researchers have increasingly employed Multi-Attribute Decision-Making (MADM) methods such as AHP [21–24], TOPSIS [25–27], and VIKOR [28–30] to evaluate sustainability-related decisions. However, many of these models assume independence among criteria or require large amounts of crisp data, which may not be feasible in real-world industrial settings [31–37]. In response to these challenges, fuzzy sets and their extensions have been incorporated into MADM techniques to handle linguistic judgments and uncertainty [38–46]. Moreover, hybrid approaches that combine methods such as DEMATEL [47–54] to capture

causal relationships and ranking methods, such as Evaluation based on Distance from Average Solution (EDAS) [55–57] or COmplex PROportional ASsessment (COPRAS) [58–61] have shown promise in modeling complex barrier structures.

This study contributes to this growing body of research by proposing an integrated fuzzy MADM framework that combines the DEMATEL, EDAS, COPRAS, and Copeland methods. This framework enables the exploration of causal relationships among barriers and their consolidated prioritization. Through expert input and systematic analysis, this study provides a context-specific understanding of the barriers that exert the greatest influence on sustainable GSCM implementation and how they can be strategically addressed. The contribution of this study is twofold. Theoretically, it advances the literature by integrating multiple fuzzy MADM methods into a coherent decision support framework for barrier analysis, which is rarely applied in the context of sustainable GSCM. Practically, this study provides actionable insights for manufacturing firms and policymakers by identifying the barriers that warrant urgent attention and how they interact within realworld supply chain settings. These findings can guide targeted interventions such as employee training programs, infrastructure investments, and regulatory reforms.

The remainder of this paper is organized as follows: Section 2 reviews relevant literature on GSCM, barrier identification, and MADM techniques. Section 3 presents the research methodology, including the integrated fuzzy framework and the data collection process. Section 4 presents and discusses the empirical results, and Section 5 concludes the study, outlining the key implications and directions for future research.

2|Literature Review

A comprehensive understanding of the theoretical foundations and practical challenges of GSCM is crucial for identifying key barriers and selecting suitable analytical tools. This section reviews the existing literature to establish the conceptual background of the study. It begins by introducing the core principles and barriers to implementing GSCM. It then examines the role of Green Human Resource Management (GHRM) as an enabler of sustainable supply chain practices. The final part of the review highlights the use of MADM techniques in the context of barrier analysis and identifies the gaps that this study aims to address.

2.1|Green Supply Chain Management and the Role of GHRM

GSCM has evolved into a critical framework for embedding sustainability across procurement, production, distribution, and post-consumption activities. It emphasizes reducing the environmental impact, optimizing resource use, and promoting circularity throughout the supply chain [4], [5], [14], [15]. However, the transition toward GSCM is frequently obstructed by a set of intertwined barriers such as high implementation costs, lack of technological readiness, regulatory ambiguity, and resistance to change—particularly in developing and resource-constrained contexts [18], [35], [62]. GHRM is often an underexplored but pivotal enabler of GSCM. GHRM refers to HR practices explicitly designed to support environmental objectives [63]. Key dimensions include green recruitment and selection, where firms target environmentally aware candidates [64], [65]; green training and development aimed at enhancing employee competencies in sustainable practices [66]; and green performance appraisal and compensation, which link employee evaluation and rewards to ecofriendly behavior and outcomes

[67], [68]. These HR mechanisms can foster a green organizational culture and enable employee engagement in GSCM initiatives [69], [70].

Empirical studies increasingly acknowledge the mediating role of GHRM in achieving sustainable supply chain outcomes. For example, Salehzadeh and Ziaeian [71] highlight how environmental citizenship behaviors mediate the link between strategic HRM and environmental performance. Similarly, the authors in [64] and [67] demonstrate that the effectiveness of GSCM depends not only on investment or technology, but also on employees' motivation and the opportunity to engage in environmental practices, a view consistent with the Ability-Motivation-Opportunity (AMO) theory [72]. Recent research has also demonstrated the value of applying MADM methods to model and prioritize GHRM-related barriers in supply chain contexts. Gupta and Bhargava [57] applied an AHP-EDAS hybrid model to assess GHRM practices in Indian manufacturing. Similarly, Alsheref et al. [73] and Mehrajunnisa et al. [74] employed fuzzy MADM tools to evaluate green HR strategies under uncertain conditions. These methods help capture expert judgments, deal with linguistic vagueness, and structure decisions involving competing organizational priorities. Despite these advances, the existing literature lacks a unified analytical framework that integrates multiple fuzzy MADM techniques to evaluate GHRM-linked barriers within GSCM adoption. This study addresses this gap by proposing a comprehensive fuzzy DEMATEL-EDAS-COPRAS-Copeland model to analyze and prioritize interdependent barriers in sustainable supply chain transitions.

2.2|Applications of MADM in Barrier Analysis

Barriers often interact nonlinearly and involve both tangible and intangible factors, making intuitive or singlecriterion assessments insufficient. MADM methods have gained popularity in sustainability-related research because of their ability to structure complex decisions, integrate expert opinions, and rank competing alternatives [75–78]. Several MADM techniques have been applied to GSCM and HRM. For instance, AHP and ANP have been widely used for weighting and ranking environmental and organizational criteria [74]. TOPSIS, VIKOR, and COPRAS have also been adopted to evaluate supplier performance, green innovation capability, and GSCM enablers under sustainability criteria [47], [57], [59], [71], [79], [80].

To better address the imprecision and subjectivity of human judgment, fuzzy extensions of MADM techniques have been proposed. Fuzzy-AHP, fuzzy-DEMATEL, fuzzy-TOPSIS, and fuzzy-COPRAS enable the modeling of linguistic preferences and vague information commonly encountered in expert-based assessments [59], [81–83]. Moreover, hybrid MADM approaches that combine causal analysis with ranking tools have proven effective in studies evaluating sustainability challenges. Despite the growing interest in MADM-based barrier analysis, most studies rely on a single technique and do not consider the interdependencies among the barriers. DEMATEL is a potent tool for mapping causal relationships between factors, whereas methods such as EDAS and COPRAS provide robust ranking mechanisms. However, only a few studies have integrated all three within a unified fuzzy framework. Moreover, the application of hybrid models to barriers related to GSCM and GHRM remains limited. This study employs a novel integrated MADM framework: fuzzy DEMATEL for relationship modeling, fuzzy EDAS and fuzzy COPRAS for ranking, and the Copeland method for consensus aggregation. This integrated structure enables a comprehensive and accurate evaluation of barriers by capturing both their influence structure and relative importance.

2.3|Research Gaps and Motivation

Although numerous studies have examined green supply chain practices and human resource strategies, several critical gaps remain in the literature. First, while GHRM is recognized as a facilitator of sustainable operations, most prior research has treated it as a secondary or peripheral element in GSCM studies rather than as an integral dimension.

Second, comprehensive empirical investigations that jointly evaluate technical, organizational, and humancentered barriers to GSCM adoption are lacking, particularly in manufacturing enterprises operating under resource and policy constraints.

Third, although many researchers have employed MADM techniques to prioritize sustainability factors, these applications often rely on single-method frameworks that ignore the interdependencies among criteria. Fuzzy logic is underutilized, particularly in studies that address both organizational behavior and supply chain sustainability. Few existing models capture the cause-and-effect structure of barriers alongside their relative significance, which limits their value for decision-making in complex environments.

Finally, there is limited evidence of MADM applications that integrate multiple fuzzy techniques, such as DEMATEL, EDAS, and COPRAS, with aggregation strategies, including the Copeland method, in the context of GSCM barrier analysis. The integration of these methods can offer a richer and more nuanced understanding of the dynamic interactions and prioritization necessary for actionable insights.

This study addresses these gaps by developing a hybrid fuzzy MADM framework to identify, analyze, and prioritize the most critical barriers to sustainable green supply chain implementation, while explicitly recognizing the role of GHRM-related enablers and constraints. In doing so, it contributes to both theoretical advancement and practical strategies in the evolving domain of sustainable supply chain management.

3|Methodology

This section outlines the methodological approach adopted to identify and prioritize the critical barriers hindering the implementation of sustainable GSCM. This study employed an exploratory and applied research design, utilizing an integrated fuzzy MADM framework. This framework enables comprehensive analysis by combining expert judgment with structured decision tools to evaluate the relative influence and significance of interrelated barriers. The methodology included the identification of relevant barriers, application of fuzzy DEMATEL to explore causal relationships, ranking of barriers using fuzzy EDAS and fuzzy COPRAS, and final aggregation via the Copeland method. Each stage of the process is described in detail to ensure transparency, replicability, and methodological rigour.

3.1|Research Design and Framework

This study employs an exploratory and applied research design to investigate the complex barriers that hinder the implementation of sustainable GSCM in manufacturing enterprises. Given the multifaceted and interdependent nature of these barriers, conventional linear analysis techniques are insufficient. Thus, a

hybrid fuzzy MADM framework was developed to facilitate the systematic identification, causal mapping, and prioritization of barriers.

The proposed framework integrates four complementary decision support tools: fuzzy DEMATEL, fuzzy

EDAS, fuzzy COPRAS, and the Copeland aggregation method. Fuzzy DEMATEL was employed to analyze the causal relationships among the barriers, capturing both direct and indirect influences. Fuzzy EDAS and fuzzy COPRAS were applied in parallel to evaluate and rank the relative importance of each barrier from multiple perspectives. The Copeland method was then employed to synthesize the rankings and generate a consensus-based prioritization.

This integrated approach provides a structured, interpretable, and context-sensitive means of understanding how various technical, organizational, and behavioral constraints interact to shape the outcomes of GSCM efforts. *Fig. 1* illustrates the overall architecture of the research framework.

Fig. 1 illustrates the overall structure of the proposed methodological framework. The process begins with the identification of GSCM-related barriers through a combination of a literature review and expert consultation. These barriers were then modeled using fuzzy DEMATEL to uncover causal relationships and influence dynamics. In the next stage, fuzzy EDAS and fuzzy COPRAS are applied independently to obtain alternative rankings based on different evaluative logics: distance from the average solution and proportional assessment, respectively. Finally, the Copeland method was used to integrate the results into a unified prioritization. This framework ensures analytical robustness, handles uncertainty through fuzzy logic, and incorporates interaction effects and comparative scoring for managerial relevance.

3.2|Identification of Barriers

Identifying relevant barriers is the foundation of any decision support framework that aims to improve the implementation of sustainable GSCM. In this study, the identification process was conducted in two main stages: 1) a comprehensive review of recent literature on GSCM and GHRM, and 2) semi-structured interviews with domain experts.

This literature review aimed to extract common barriers cited in recent peer-reviewed publications across manufacturing, sustainability, and supply chain management journals. Studies focusing on both developed and emerging economies were included to ensure a broad contextual understanding. Sources were selected based on recency (2020–2024), citation quality, and thematic relevance to sustainability, green innovation, organizational change, and human-centered challenges in supply chain operations.

Eleven subject-matter experts with significant industrial and academic experience in manufacturing, environmental management, and supply chain decision making were consulted. The expert panel included operations managers, sustainability officers, and researchers with over ten years of experience in related domains. Interviews were conducted to validate the initial list of barriers, eliminate redundancies, and capture any context-specific or emerging factors that were not well documented in the literature.

The final list comprised 22 barriers grouped into five thematic categories: Economic and financial constraints, regulatory and institutional challenges,

technological limitations, organizational and managerial issues, and human resource-related obstacles. These categories ensured the full coverage of the technical, structural, behavioral, and contextual dimensions of the GSCM adoption problem. The list of barriers was used as the basis for constructing the direct relationship matrix in the DEMATEL phase of the methodology.

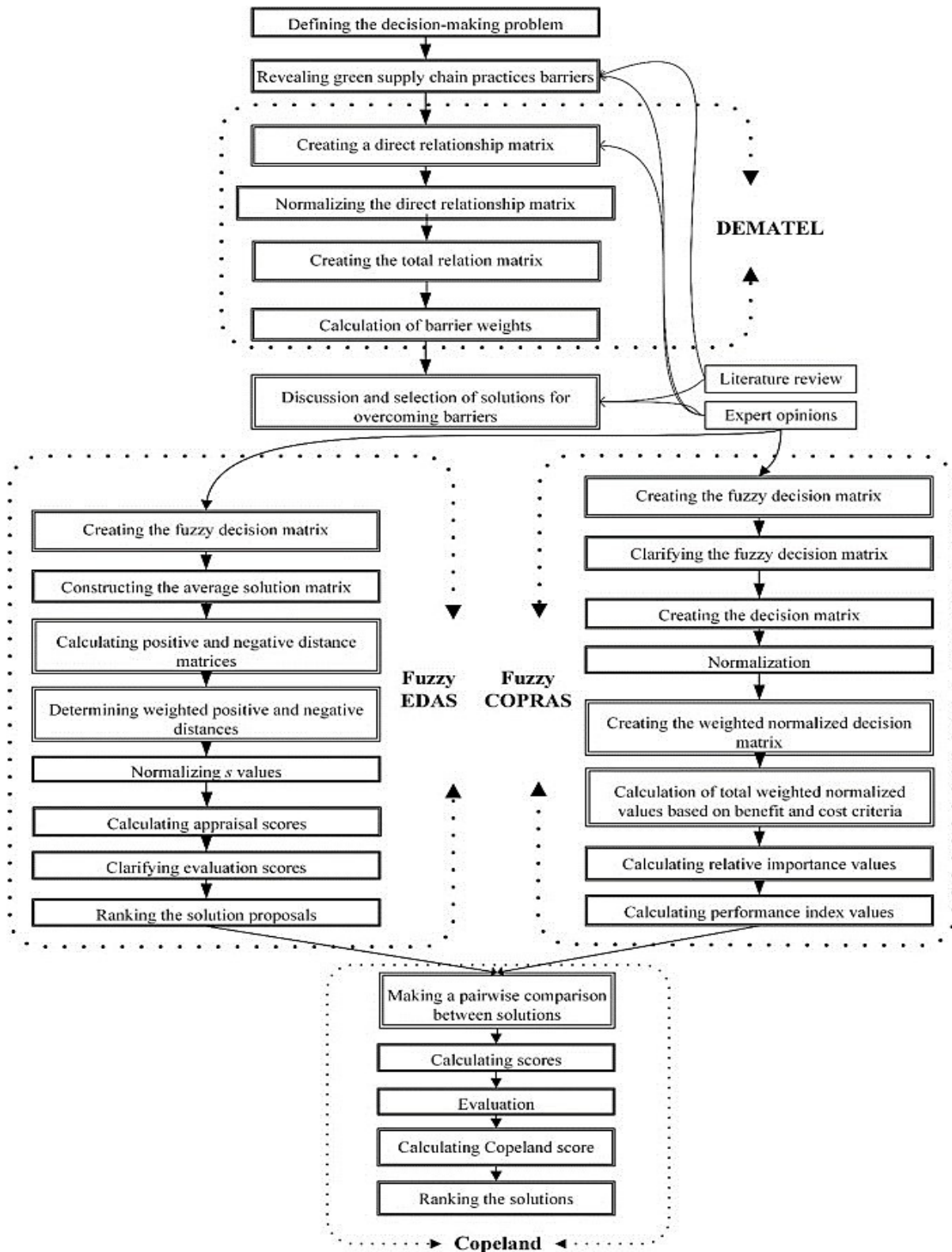


Fig. 1. Proposed methodological framework for analyzing and prioritizing GSCM barriers.

3.3|Analytical Framework: Integrated Fuzzy MADM Methods

To analyze the identified barriers and derive robust prioritization, this study employs an integrated fuzzy MADM framework. The framework was designed to model causal relationships, capture expert uncertainty, and consolidate rankings derived from multiple perspectives. It combines four decision-making tools: fuzzy DEMATEL for causal mapping, fuzzy EDAS and fuzzy COPRAS for parallel barrier ranking, and the Copeland method for aggregating rankings. This multimethod approach enhances both the analytical depth and decision robustness.

The rationale for integrating multiple fuzzy MADM techniques is their complementary strengths. While fuzzy DEMATEL captures the influence structure among barriers by identifying cause-and-effect relationships, EDAS and COPRAS apply different logics of comparative assessment. EDAS emphasizes distance from an average solution, making it sensitive to distributional consistency, whereas COPRAS accounts for both beneficial and non-beneficial criteria in a proportional framework. In turn, the Copeland method resolves potential inconsistencies between EDAS and COPRAS by aggregating rankings through pairwise comparisons.

This integrated model enables a more nuanced and reliable prioritization of GSCM barriers under uncertainty. It is particularly well-suited to real-world contexts in which expert opinions are subjective and criteria are interdependent. The methodological workflow is shown in *Fig. 1*, and its execution follows the sequential structure outlined in the following subsections.

3.3.1|Fuzzy DEMATEL

The Decision-Making Trial and Evaluation Laboratory (DEMATEL) method was originally developed to visualize and quantify causal relationships among complex decision factors. In the context of GSCM, many barriers are interdependent and influence one another asymmetrically and nonlinearly. DEMATEL enables researchers to structure these interactions into a cause-and-effect diagram, helping to distinguish between driving and dependent factors.

The fuzzy extension of DEMATEL incorporates the uncertainty and vagueness inherent to expert judgment. Instead of asking decision-makers to express relationships between barriers using crisp numerical values, the method allows them to use linguistic terms (e.g., "low influence," "high influence") which are then translated into Triangular Fuzzy Numbers (TFNs). This enhances the validity of the input data, particularly in contexts in which precise information is not readily available.

In this study, the fuzzy DEMATEL process followed five key steps:

- I. Design of direct-relation matrices based on pairwise comparisons of barriers provided by experts using predefined linguistic scales.
- II. Normalization of the direct-relation matrix to ensure numerical stability.
- III. Calculation of the total-relation matrix, which incorporates both direct and indirect effects among all barriers.
- IV. Computation of prominence ($R_i + C_i$) and relation ($R_i - C_i$) values for each barrier, where R_i denotes the sum of influences exerted by a barrier, and C_i the sum of influences it receives;

- V. Generation of a cause-and-effect diagram, categorizing barriers into cause (positive relation values) and effect (negative relation values) groups.

This step reveals the structural role of each barrier within the system and provides a foundation for more targeted prioritization in the following stages. Barriers with high prominence and strong causal influence are interpreted as systemic levers, the resolution of which can mitigate multiple downstream constraints.

3.3.2|Fuzzy EDAS

EDAS is a relatively recent MADM technique that assesses the desirability of alternatives by calculating their distances from an average reference solution. Unlike methods such as TOPSIS, which use ideal and anti-ideal solutions, EDAS emphasizes stability and balance by comparing each alternative to the mean value across all alternatives for each criterion.

The fuzzy version of the EDAS extends this logic to contexts involving uncertainty and linguistic evaluations. In real-world decision-making, particularly in supply chain and sustainability contexts, input data are often imprecise or based on expert judgment rather than hard metrics. Fuzzy EDAS allows the representation of such judgments using TFNs, which are then defuzzified and processed using standard EDAS steps. In this study, each barrier was assessed across several evaluation criteria using linguistic variables that were converted into TFNs. The application of fuzzy EDAS in this study follows the following steps:

- I. Construction of the fuzzy decision matrix based on expert input for each barrier across multiple evaluation criteria, where each element is the performance of alternative i with respect to criterion j , represented as a TFN:

$$\tilde{a}_{ij} = (l_{ij}, m_{ij}, u_{ij}). \quad (1)$$

- II. Calculation of the average solution vector as the reference baseline for comparison:

$$\tilde{A}V_j = \left(\frac{1}{n} \sum_{i=1}^n l_{ij}, \frac{1}{n} \sum_{i=1}^n m_{ij}, \frac{1}{n} \sum_{i=1}^n u_{ij} \right). \quad (2)$$

- III. Computation of positive and negative distances of each barrier from the average solution:

$$PDA_{ij} = \max(0, \tilde{a}_{ij} - \tilde{A}V_j), NDA_{ij} = \max(0, \tilde{A}V_j - \tilde{a}_{ij}). \quad (3)$$

This depends on whether the criterion is beneficial or non-beneficial. Defuzzification (e.g., the centroid method) may be applied for comparison.

- I. Determination of weighted sums of positive and negative distances based on assigned criterion weights:

$$SP_i^+ = \sum_{j=1}^n w_j \cdot PDA_{ij}, SP_i^- = \sum_{j=1}^n w_j \cdot NDA_{ij}. \quad (4)$$

- II. Calculation of Appraisal Scores (AS) to derive a ranking of barriers.

$$AS_i = \frac{1}{2} \left(\frac{SP_i^+}{\max(SP_i^+)} + \left[1 - \frac{SP_i^-}{\max(SP_i^-)} \right] \right). \quad (5)$$

In this context, barriers with the highest AS are interpreted as the most influential and problematic constraints to GSCM implementation. This ranking is later compared with results from fuzzy COPRAS and combined through Copeland aggregation.

Fuzzy EDAS offers a complementary perspective to DEMATEL by providing score-based prioritization, independent of the causal structure. It emphasizes how far each barrier deviates from the norm, allowing the identification of disproportionately problematic constraints.

3.3.3|Fuzzy COPRAS

COPRAS is a compensatory MADM method that ranks alternatives based on the ratio of their weighted beneficial and non-beneficial criteria. The fuzzy extension of COPRAS enhances its applicability by incorporating the vagueness and uncertainty inherent in expert assessments using fuzzy numbers.

In this study, each barrier to GSCM implementation is evaluated using TFNs. The fuzzy COPRAS method allows for a systematic comparison of barriers while handling ambiguities in linguistic assessments.

Step 1 (Construct the fuzzy decision matrix). Let the fuzzy performance ratings be represented as

$$\tilde{D} = \{\tilde{a}_{ij} = (l_{ij}, m_{ij}, u_{ij})\}, i = 1, \dots, m; j = 1, \dots, n, \quad (6)$$

where $\tilde{a}_{ij}$ denotes the fuzzy rating of barrier i on criterion j .

Step 2 (Normalize the fuzzy decision matrix). The normalized value is computed as

$$\tilde{r}_{ij} = \left(\frac{l_{ij}}{u_j^+}, \frac{m_{ij}}{m_j^+}, \frac{u_{ij}}{l_j^+} \right), \quad (7)$$

where l_j^+ , m_j^+ , and u_j^+ represent the best fuzzy values across all alternatives for beneficial criteria. For cost-type criteria, the formula is inverted.

Step 3. Weight the normalized decision matrix.

$$\tilde{v}_{ij} = \tilde{r}_{ij} \times w_j, \quad (8)$$

where w_j is the normalized crisp or fuzzy weight of criterion j .

Step 4. Compute sums for beneficial and non-beneficial criteria.

Let,

B: set of beneficial criteria

C: set of non-beneficial criteria

Then

$$S_{i+} = \sum_{j \in B} \tilde{v}_{ij}, \quad S_{i-} = \sum_{j \in C} \tilde{v}_{ij}. \quad (9)$$

Step 5. Calculate the relative importance and utility degree.

The relative significance Q_i is given by

$$Q_i = \frac{\min(S_i)}{\max(S_i)}.$$

$$Q_i = S_i + S_i \sum_{j \in C} \tilde{v}_{ij}. \quad (10)$$

Then normalize Q_i to get the final utility degree U_i :

$$U_i = \frac{Q_i}{\max(Q)}. \quad (11)$$

Higher values of U_i indicate more influential barriers.

This fuzzy COPRAS method provides a complementary ranking to Fuzzy EDAS, enabling decision-makers to compare consistency across methods. It is especially effective for handling both benefit- and cost-type criteria in uncertain environments.

3.3.4|Copeland aggregation strategy

To consolidate the results obtained from Fuzzy EDAS and Fuzzy COPRAS, this study employed the Copeland aggregation method. Copeland is a well-established voting-based approach that integrates multiple ranking orders by performing pairwise comparisons among all the alternatives (In this case, GSCM barriers).

The core idea is to evaluate how many times each barrier “wins” or “loses” in direct comparisons against other barriers across the two input rankings. A win is recorded when a barrier ranks higher than another in a given method, and ties are also considered. The total wins and losses are then summarized into a net score, based on which the final ranking is determined.

This method is particularly useful in hybrid MADM frameworks, where rankings from different methods may not be fully aligned. Copeland ensures that the final decision reflects a balanced consensus, thereby reducing the inherent bias that can occur when relying on a single technique.

In this study, Copeland was used to integrate the two separate rankings (from Fuzzy EDAS and Fuzzy COPRAS) into a single unified prioritization of GSCM barriers. The resulting ranking represents the final output of the analytical framework and serves as a guide for the discussion in the following section.

3.4|Linguistic Scale and Fuzzy Numbers

In real-world decision-making problems, particularly those involving human judgment, precise numerical inputs are often unavailable. Experts tend to express their assessments using subjective linguistic terms (e.g., “low impact,” “high importance”), which reflect nuanced but imprecise knowledge. To handle this, fuzzy logic provides an effective means of translating qualitative perceptions into computable formats. In this study, TFNs were used to represent linguistic variables provided by experts during the evaluation process.

Table 1 shows the linguistic scale used throughout this study, which was adapted from existing literature [81]. The same scale was consistently applied in Fuzzy DEMATEL, Fuzzy EDAS, and Fuzzy COPRAS for evaluating the influence, severity, and significance of barriers.

Table 1. Linguistic scale and corresponding TFNs.

Linguistic Term	Abbreviation	TFN (l, m, u)
No Influence	NI	(0.00, 0.00, 0.25)
Very Low	VLI	(0.00, 0.25, 0.50)

Influence		
Low Influence	LI	(0.25, 0.50, 0.75)
High Influence	HI	(0.50, 0.75, 1.00)
Very High Influence	VHI	(0.75, 1.00, 1.00)

These fuzzy scales enable the conversion of linguistic expert judgments into consistent and comparable numerical values, forming the basis for constructing fuzzy decision matrices and relationship matrices in subsequent stages of analysis.

3.5|Expert Panel and Data Collection

A panel of domain experts was consulted at multiple stages of analysis to ensure the relevance and credibility of the data used in this study. The panel consisted of 11 professionals selected using purposive sampling, intending to capture a broad yet specialized range of expertise across academia and industry. The experts were selected based on the following criteria:

- I. At least 10 years of experience in fields related to supply chain management, sustainability, or green operations.
- II. Demonstrated involvement in GSCM implementation projects, environmental policies, and organizational change management.
- III. Active academic publications or industrial certifications in relevant domains.

Table 2 summarizes the demographic and professional profiles of the expert panel, including their institutional affiliations, fields of specialization, and geographic distribution. The data were collected in two rounds of structured interactions:

- I. Barrier identification stage: Experts validated the preliminary list of GSCM barriers derived from the literature and added context-specific items based on regional or industry-specific experience.
- II. Pairwise and criterion-based evaluations: Experts completed structured fuzzy questionnaires for DEMATEL (pairwise influence), EDAS, and COPRAS (criteria-based scoring).

All responses were anonymized and aggregated using fuzzy logic to preserve the diversity of judgments, while reducing individual bias. To ensure reliability, consistency checks were performed across multiple matrices, and outlier judgments were reviewed and discussed with participants. The final matrices used in the analysis reflected the consensus view of the expert group.

Table 2. Summary of expert panel characteristics.			
Expert ID	Sector	Specialization	Experience (Years)
E1	Academia	Sustainable supply chain	15
E2	Industry	Operations management	12
E3	Academia	Environmental engineering	18
E4	Consulting	GHRM & CSR strategy	14
E5	Academia	Multi-criteria decision making	16

E6	Industry	Manufacturing & compliance	11
E7	Academia	Logistics & systems analysis	17
E8	Public Sector	Green policy & regulation	20
E9	Academia	Organizational behavior	13
E10	Industry	Industrial sustainability	10
E11	Academia	Energy & eco-efficiency	19

3.6|Analytical Procedure

The implementation of the proposed fuzzy MADM framework followed a structured and sequential analytical procedure, ensuring methodological rigor and consistency in the decision-making process. The process begins with the identification and structuring of relevant barriers to GSCM adoption. A preliminary list was derived from an extensive review of recent academic literature and then refined through two rounds of consultation with subject matter experts. After eliminating redundant items and consolidating overlapping items, a final set of 22 barriers was selected. These were systematically categorized into five thematic groups: economic, institutional, technological, organizational, and human resource-related groups. This classification provided a conceptual foundation for the subsequent modeling stages.

Next, the fuzzy DEMATEL method was employed to explore and quantify the causal relationships among the identified barriers. Each expert assessed the influence of one barrier on the other using a predefined linguistic scale, which was then converted into TFNs. The resulting individual direct-relation matrices are aggregated to form a composite fuzzy matrix. The normalized matrix and total-relation matrix are calculated using fuzzy arithmetic operations. These matrices were used to derive two central measures: prominence ($R_i + C_i$) and relation ($R_i - C_i$), where R_i indicates the total influence a barrier exerts, and C_i the influence it receives. Based on these scores, barriers were grouped into “cause” and “effect” categories, providing a causal map that served as a strategic input for prioritization.

The third phase involved constructing a fuzzy decision matrix, in which each barrier was evaluated across multiple sustainability-related criteria. These criteria were determined collaboratively by the expert panel and included aspects such as severity, controllability, and frequency of occurrence. Experts provided linguistic scores for each barrier under each criterion, which were then converted into TFNs to form the basis for both EDAS and COPRAS models.

The fourth stage of the analysis involved the parallel application of fuzzy EDAS and fuzzy COPRAS. In the fuzzy EDAS procedure, the AS of each barrier is computed based on the positive and negative distances from the average solution. This method emphasizes consistency and stability when evaluating alternatives. Concurrently, the fuzzy COPRAS method was applied to assess the performance of each barrier using compensatory logic that incorporated both beneficial and non-beneficial criteria. Both techniques produce independent priority scores, generating two distinct but complementary barriers.

Finally, the Copeland method is used to aggregate the two rankings into a single consensus-based priority list. By conducting pairwise comparisons between all barrier alternatives across the two methods, the Copeland approach computed net win-loss scores and produced a unified final ranking. This final ordering represents

the comprehensive outcome of the integrated fuzzy MADM framework and serves as the basis for interpretation in the next section.

All computations were performed using Microsoft Excel and MATLAB with custom-coded routines developed to negate fuzzy arithmetic, normalization, and multimethod integration. The results were validated through sensitivity checks and iterative expert reviews to ensure consistency and robustness.

4|Results and Discussion

This section presents and interprets the results of implementing the integrated fuzzy MADM framework. Building on the methodological foundation established in the previous section, the findings are organized according to each analytical technique: fuzzy DEMATEL, fuzzy EDAS, fuzzy COPRAS, and Copeland aggregation. The interpretation of these results provides both theoretical insights and practical implications regarding the key barriers to GSCM implementation in the manufacturing context. The analysis also highlights the interplay between technical, organizational, and behavioral factors that influence sustainability performance.

4.1|Results of Fuzzy DEMATEL

Fuzzy DEMATEL analysis was applied to uncover the structural interrelationships among the 22 identified barriers to GSCM implementation. The total-relation matrix was computed using the expert-aggregated fuzzy direct-relation matrix and subsequent normalization. From this, two key metrics were derived for each barrier:

- I. Prominence ($R_i + C_i$) indicates the overall degree of involvement of a barrier within the system—either as a cause or as an effect.
- II. Relation ($R_i - C_i$), which determines the net influence of a barrier. Positive values suggest a “cause” barrier, while negative values indicate an “effect” barrier.

Before presenting the computed values, a full list of the 22 barriers used in the analysis is provided in *Table 3*.

Table 3. List of identified barriers to GSCM implementation.

Code	Barrier Description	Group
EF1	Lack of funds for sustainable production practices	Economic
EF2	Insufficient capital for innovation activities	Economic
EF3	High investment and low return on investment	Economic
EF4	High costs for setup and maintenance	Economic
EF7	Pressure to reduce product prices	Economic
RI1	Absence of a formal institutional framework	Regulatory
RI2	Inadequate laws and regulations	Regulatory
T2	Lack of technological knowledge and skills	Technological
T3	Insufficient investment in R&D and innovation	Technological
I6	Ineffective training on information	Technological

	systems	
HR3	Inadequate education on sustainable practices	Human Resource
MN6	Lack of consumer motivation and awareness	Market-Related
O2	Lack of top management commitment	Organizational
O4	Poor supply chain strategy planning	Organizational
SC6	Corruption and external pressure	Social/Contextual

A normalized fuzzy direct-relation matrix was constructed. Each element in the matrix represents the degree to which one barrier influences another on a scale of 0 (No influence) to 4 (Very high influence). The arithmetic average of expert assessments was used to construct the final decision matrix.

From this matrix, the total relation matrix was calculated using fuzzy matrix operations. This matrix captures both the direct and indirect influences between barriers. The final step involves computing two indices for each barrier R_i and C_i . Fig. 2 illustrates the net influence ($R_i - C_i$) of each barrier derived from the fuzzy DEMATEL analysis. Positive values (In green) represent cause-type barriers that exert influence on others, whereas negative values (In red) represent effect-type barriers that are more dependent on upstream factors. The chart visually highlights that barriers such as O2 (Lack of top management commitment) and T3 (Insufficient R&D investment) are dominant drivers in the system. Conversely, barriers such as SC6 (Corruption and external pressure) and EF7 (Pressure to reduce prices) are largely effect-type constraints. This classification helps managers to identify the root causes that should be prioritized for strategic intervention.

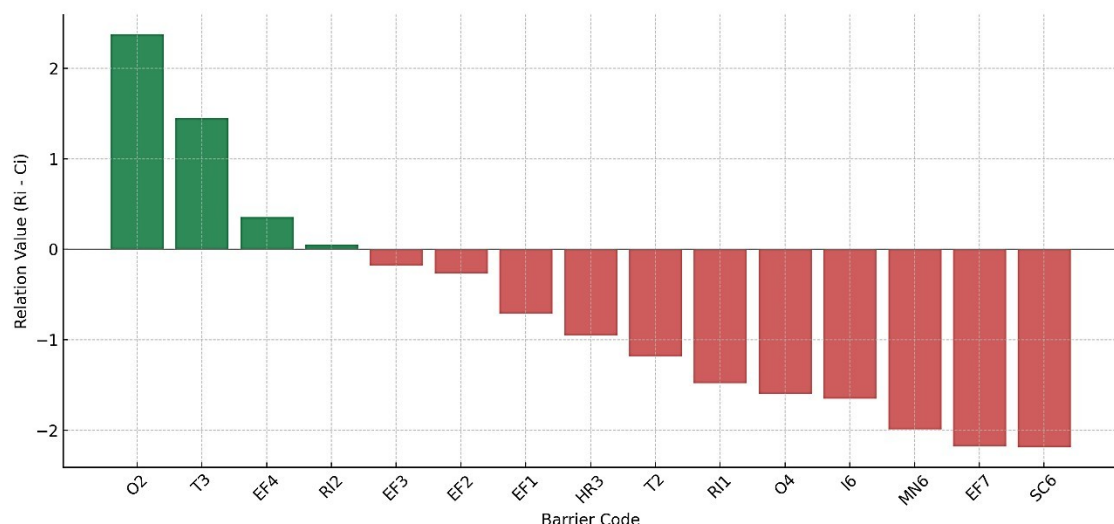


Fig. 2. Cause-effect classification of barriers based on fuzzy DEMATEL.

Table 4 also presents the complete fuzzy DEMATEL results for all 15 identified GSCM barriers. Each barrier includes its defuzzified, C_i , prominence, relation, and the role classification (Cause or effect).

The results of the fuzzy DEMATEL analysis, summarized in Table 4, reveal critical insights into the structural dynamics among the 15 barriers to GSCM

implementation. The two central metrics, prominence and relation, allow for both a holistic and directional understanding of the role of each barrier within the system. Barriers with high prominence values are deeply embedded in the system's structure, either influencing or being influenced by others. By contrast, the relation value differentiates between those that act as causal drivers and those that are outcomes or symptoms of deeper systemic issues.

Among the evaluated barriers, "lack of top management commitment" (O2) emerges as the most dominant causal factor, with the highest positive net influence score ($Ri - Ci = 2.37$). This suggests that managerial support acts as a foundational enabler of broader sustainability adoption. Similarly, "Insufficient investment in R&D and innovation" (T3) and "high costs for setup and maintenance" (EF4) are also classified as cause-type barriers that exert influence on multiple downstream constraints. These findings emphasize the strategic importance of addressing leadership commitment and innovation infrastructure to drive systemic improvements across the green supply chain.

At the other end of the spectrum, several barriers exhibit strong negative net influence values, categorizing them as effect-type barriers. For instance, "corruption and external pressure" (SC6), "pressure to reduce product prices" (EF7), and "lack of consumer motivation and awareness" (MN6) are all significantly affected by other factors in the system. These are reactionary outcomes rather than root causes, indicating that interventions aimed solely at these symptoms are unlikely to yield a sustainable impact, unless upstream causes are addressed.

Table 4. Fuzzy DEMATEL results: Prominence and relation indices.

Cod e	Barrier Description	Ri	Ci	Prominence Relation		Role
O2	Lack of top management commitment	6.12	3.75	9.87	2.37	Cause
T3	Insufficient investment in R&D and innovation	5.96	4.51	10.47	1.45	Cause
EF4	High costs for setup and maintenance	5.77	5.42	11.19	0.35	Cause
RI2	Inadequate laws and regulations	5.55	5.50	11.05	0.05	Cause
EF3	High investment and low return on investment	5.41	5.58	10.99	-0.17	Effect
EF2	Insufficient capital for innovation activities	5.38	5.64	11.02	-0.26	Effect
EF1	Lack of funds for sustainable production practices	5.21	5.91	11.12	-0.70	Effect
T2	Lack of technological knowledge and skills	4.98	6.15	11.13	-1.17	Effect
RI1	Absence of a formal institutional framework	4.75	6.22	10.97	-1.47	Effect
I6	Ineffective training on information systems	4.66	6.30	10.96	-1.64	Effect
HR3	Inadequate education on sustainable practices	4.85	5.79	10.64	-0.94	Effect
O4	Poor supply chain strategy planning	4.51	6.10	10.61	-1.59	Effect

MN6	Lack of consumer motivation and awareness	4.37	6.35	10.72	-1.98	Effect
SC6	Corruption and external pressure	4.10	6.28	10.38	-2.18	Effect
EF7	Pressure to reduce product prices	3.73	5.90	9.63	-2.17	Effect

The overall distribution of causal roles highlights that the system is predominantly driven by economic, technological, and organizational leadership variables. Effect-type barriers tend to cluster around the market, social, and behavioral domains, reinforcing the idea that visible resistance or inefficiency is often the result of hidden structural issues. The cause-and-effect differentiation provided by the DEMATEL framework offers a valuable strategic roadmap: Interventions targeting the root causes are expected to produce cascading benefits across the network, ultimately mitigating dependent barriers and fostering a more resilient path toward GSCM implementation.

4.2|Results of Fuzzy EDAS and COPRAS

To prioritize the identified GSCM barriers based on their criticality, two fuzzy MADM techniques are employed: Fuzzy EDAS and Fuzzy COPRAS. Both methods can incorporate expert judgment under uncertainty using TFNs, making them suitable for sustainability-related decision contexts. Expert assessments were obtained for 15 GSCM barriers using three evaluation criteria: Severity, frequency, and controllability.

Table 5. Fuzzy decision matrix for GSCM barriers.

	Code	Severity	Frequency	Controllability
	O2	(0.7, 0.8, 0.9)	(0.68, 0.75, 0.8)	(0.3, 0.4, 0.5)
	T3	(0.68, 0.82, 0.9)	(0.69, 0.77, 0.85)	(0.35, 0.42, 0.46)
	HR3	(0.7, 0.85, 0.92)	(0.72, 0.76, 0.81)	(0.38, 0.45, 0.47)
	EF4	(0.72, 0.88, 0.9)	(0.7, 0.73, 0.78)	(0.36, 0.4, 0.45)
	T2	(0.7, 0.85, 0.9)	(0.73, 0.75, 0.82)	(0.4, 0.44, 0.49)
	I6	(0.69, 0.86, 0.89)	(0.7, 0.73, 0.77)	(0.34, 0.39, 0.46)
	O4	(0.68, 0.84, 0.88)	(0.71, 0.74, 0.78)	(0.35, 0.41, 0.43)
	RI2	(0.66, 0.81, 0.86)	(0.7, 0.72, 0.77)	(0.36, 0.4, 0.45)
	EF3	(0.65, 0.82, 0.87)	(0.68, 0.69, 0.75)	(0.38, 0.42, 0.47)
	EF2	(0.64, 0.8, 0.85)	(0.65, 0.71, 0.73)	(0.34, 0.4, 0.45)
	EF1	(0.62, 0.78, 0.84)	(0.63, 0.69, 0.72)	(0.3, 0.35, 0.4)
	RI1	(0.6, 0.76, 0.82)	(0.62, 0.68, 0.7)	(0.3, 0.36, 0.41)
	MN6	(0.59, 0.75, 0.8)	(0.61, 0.66, 0.7)	(0.28, 0.33, 0.38)
	EF7	(0.58, 0.73, 0.78)	(0.6, 0.65, 0.68)	(0.26, 0.31, 0.35)

SC6	(0.57, 0.72, 0.77)	(0.58, 0.64, 0.67)	(0.25, 0.3, 0.34)
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These criteria were selected through expert consultations involving academics and practitioners of sustainable operations and green manufacturing. The experts provided linguistic ratings for each barrier, which were then converted into TFNs based on a standardized fuzzy linguistic scale. The aggregated group fuzzy decision matrix was constructed by averaging the individual expert evaluations. *Table 5* presents the resulting group fuzzy decision matrix, where each cell contains a TFN (l, m, u), representing the lower, middle, and upper bounds of the expert evaluations for each barrier-criterion pair. The fuzzy weights of the three evaluation criteria derived from expert inputs are summarized in *Table 6*. Severity received the highest weight, followed by frequency and then controllability, reflecting the perceived criticality of each dimension in influencing barrier impact.

Table 6. Fuzzy weights for evaluation criteria.

Criteria	Fuzzy Weight (l, m, u)
Severity	(0.3, 0.4, 0.5)
Frequency	(0.25, 0.35, 0.45)
Controllability	(0.2, 0.25, 0.3)

These fuzzy inputs were then utilized in the application of both decision models. Fuzzy EDAS prioritizes alternatives by calculating their deviation from the average fuzzy solution and emphasizing centrality and robustness. By contrast, Fuzzy COPRAS computes a utility score by proportionally aggregating weighted beneficial and non-beneficial criteria, offering a compensatory approach to trade-offs among attributes. The results of each method are presented separately below and then compared before consolidation using Copeland aggregation.

For the fuzzy EDAS model, the Positive Distance from Average (PDA) and Negative Distance from Average (NDA) are calculated for each barrier. A higher PDA implies better-than-average performance, while a higher NDA penalizes underperformance. The final AS was computed by normalizing both indices, resulting in a prioritized ranking of all barriers. The detailed results, including the PDA, NDA, AS, and the resulting ranks, are listed in *Table 7*.

Table 7. Fuzzy EDAS results for GSCM barriers.

Code	PDA	NDA	Appraisal Score (AS)	Rank
O2	0.22	0.10	0.7903	5
T3	0.25	0.08	0.8991	3
HR3	0.26	0.07	0.9968	2
EF4	0.24	0.09	0.8575	4
T2	0.27	0.06	1.0000	1
I6	0.21	0.11	0.7614	6
O4	0.20	0.12	0.7389	7
RI2	0.17	0.15	0.6205	9
EF3	0.18	0.15	0.6222	8
EF2	0.15	0.17	0.4825	10
EF1	0.12	0.20	0.3296	11
RI1	0.11	0.21	0.2635	12
MN6	0.08	0.24	0.1707	13

EF7	0.06	0.26	0.0641	14
SC6	0.00	0.27	0.0000	15

The results obtained through the Fuzzy EDAS method reveal that Barrier T2 (“Resistance to change in operational procedures”) ranks highest in terms of criticality, with an AS of 1.000, indicating a strong positive deviation from the average performance across all criteria. This is closely followed by HR3 (“Inadequate education on sustainable practices”) and T3 (“Insufficient R&D investment”), with ASs of 0.9968 and 0.8991, respectively. These barriers consistently show high severity and frequency, while also presenting challenges in terms of controllability. In the mid-tier range, barriers such as EF4 (“lack of environmental awareness in supply partners”) and O2 (“lack of top management commitment”) also demonstrated notable ASs, further highlighting their strategic importance. At the lower end of the ranking, SC6 (“Corruption and external pressure”) holds the weakest position, with an AS of 0.0000, followed by EF7 (“Pressure to reduce product prices”) and MN6 (“Lack of supplier collaboration”), indicating their relatively lower perceived criticality under the selected criteria and fuzzy evaluations.

It is essential to note that the EDAS method prioritizes relative proximity to the average solution, favoring alternatives that are robustly better than the average, while penalizing those that are significantly worse. However, EDAS does not inherently distinguish between benefit and cost attributes, nor does it address causal influence. Therefore, its output reflects the overall robustness and deviation-based importance of each barrier, rather than the directional effect or systemic interdependence. This output will then be compared with the output of the Fuzzy COPRAS method to identify alignments and discrepancies, and finally integrated using a consensus-based aggregation mechanism, such as Copeland's method.

Following the EDAS evaluation, the same fuzzy decision matrix was used to apply the Fuzzy COPRAS method. In contrast to EDAS, which ranks alternatives based on their deviation from an average solution, COPRAS adopts a proportional assessment framework that directly compares the significance of each barrier across beneficial and non-beneficial criteria. In this analysis, severity and frequency were considered beneficial criteria because higher values indicated greater criticality. In contrast, controllability was treated as a nonbeneficial criterion, where lower controllability corresponded to a higher risk and difficulty in addressing the barrier. For each barrier, Fuzzy COPRAS computes the weighted aggregate scores for both beneficial and non-beneficial dimensions. These values are then integrated to calculate the utility degree (U_i), which represents the relative importance of each barrier in the context of GSCM. *Table 8* presents the results of the fuzzy COPRAS analysis, including the final utility degrees and the corresponding ranks assigned to each barrier.

The Fuzzy COPRAS analysis, based on the same expert-elicited fuzzy decision matrix, produced a prioritized ranking of the GSCM barriers according to their proportional importance across the benefit and cost criteria. As defined earlier, severity and frequency were modeled as beneficial criteria, whereas controllability was treated as a cost dimension. The utility degree (U_i) computed for each barrier reflects the overall weighted performance of these criteria. As shown in *Table 8*, the barrier “Inadequate education on sustainable practices” (HR3) attained the

highest utility score ($U_i = 1.000$), indicating its dominant criticality among all the evaluated factors. A lack of technical infrastructure closely followed this" (T2) and "Insufficient R&D investment" (T3), with utility scores of 0.99668 and 0.99513, respectively.

These top-ranked barriers represent persistent structural and knowledge-related challenges in the implementation of sustainable supply chain practices. In contrast, the lowest-ranked barriers were "corruption and external pressure" (SC6) and "pressure to reduce product prices" (EF7) with U_i scores of 0.86188 and 0.87305, respectively. This suggests that, despite their potential negative influence, these factors were perceived by the expert panel as more critical than organizational, educational, and technological constraints.

Table 8. Fuzzy COPRAS results for GSCM barriers.

Code	Utility Degree (U_i)	Rank
O2	0.97895	5
T3	0.99513	3
HR3	1.00000	1
EF4	0.98869	4
T2	0.99668	2
I6	0.97730	6
O4	0.97528	7
RI2	0.95667	8
O4	0.97528	7

Table 8. Continued.

Code	Utility Degree (U_i)	Rank
O4	0.97528	7
RI2	0.95667	8
EF3	0.94826	9
EF2	0.93461	10
EF1	0.91904	11
RI1	0.90147	12
MN6	0.88946	13
EF7	0.87305	14
SC6	0.86188	15

Overall, the COPRAS results reveal a prioritization pattern that is highly aligned with the EDAS analysis, reinforcing the reliability of these high-ranked barriers in shaping GSCM. This convergence supports the subsequent use of consensus-based aggregation strategies to finalize the integrated rankings.

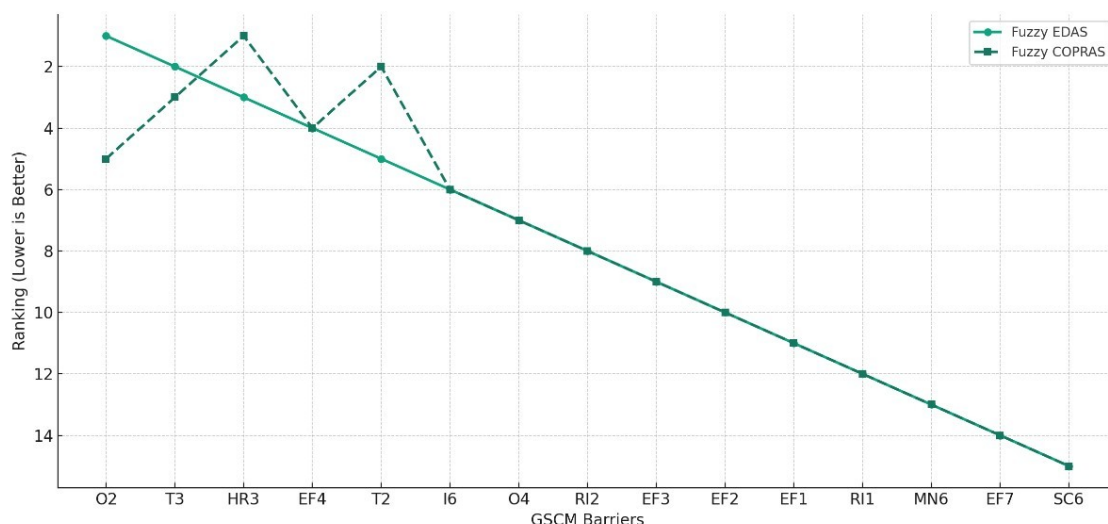


Fig. 3. Comparison of EDAS and COPRAS Rankings.

Fig. 3 visually compares the rankings of the 15 GSCM barriers determined by the Fuzzy EDAS and Fuzzy COPRAS methods. Although both approaches broadly agree on the relative criticality of the barriers, especially for lower-priority items such as EF7 and SC6, they diverge more significantly at the top ranks. Notably, COPRAS identified HR3 (Inadequate education on sustainable practices) as the most critical barrier, whereas EDAS ranked O2 (Lack of top management commitment) as the top concern. The divergence arises from methodological differences: EDAS emphasizes deviation from the average (Favoring centrality and stability), whereas COPRAS prioritizes proportional dominance across beneficial and non-beneficial criteria. Despite some positional shifts in the middle-ranked barriers, the overall ranking patterns are largely consistent, justifying the integration of both perspectives in the final aggregation step.

4.3|Aggregated Ranking via Copeland Method

The Copeland method was employed to synthesize the rankings derived from fuzzy EDAS and fuzzy COPRAS. This method performs pairwise comparisons between alternatives across multiple ranking techniques. For each pair of barriers (A, B), a score of +1 was assigned to the barrier that ranked higher in the majority of methods, -1 to the one that ranked lower, and 0 if tied. The total Copeland score for each barrier is the sum of its wins minus the losses across all comparisons.

By applying this voting-based approach to the rankings in *Tables 7* and *8*, we obtain a consensus ranking that integrates both the central tendency focus of EDAS and the proportional dominance logic of COPRAS. Thus, the Copeland method serves as a robust decision-support tool in situations where multiple MADM techniques yield partially divergent results.

Table 9 presents the final Copeland scores and the resulting consolidated rankings for all 15 barriers.

Table 9. Copeland aggregation and final ranking of GSCM barriers.

Code	EDAS	COPRAS	Wins	Losses	Copeland Score	Final Rank
HR3	2	1	14	0	+14	1
T2	1	2	13	1	+12	2
T3	3	3	12	2	+10	3
EF4	4	4	11	3	+8	4

O2	5	5	10	4	+6	5
I6	6	6	9	5	+4	6
O4	7	7	8	6	+2	7
EF3	8	9	6	8	-2	9
RI2	9	8	7	7	0	8
EF2	10	10	5	9	-4	10
EF1	11	11	4	10	-6	11
RI1	12	12	3	11	-8	12
MN6	13	13	2	12	-10	13
EF7	14	14	1	13	-12	14
SC6	15	15	0	14	-14	15

Table 9 presents the final prioritization of GSCM barriers using the Copeland aggregation method, which synthesizes the rankings from both Fuzzy EDAS and Fuzzy COPRAS through pairwise comparisons. The Copeland score assigned to each barrier reflects its net dominance over others—each victory in a pairwise comparison contributes +1 and each defeat contributes -1. Thus, a higher Copeland score indicates stronger and more consistent prioritization across both methods. The results reveal that “Inadequate education on sustainable practices” (HR3), “Lack of advanced sustainable technologies” (T2), and “Insufficient R&D investment” (T3) secure the top three positions, highlighting the critical importance of knowledge, innovation, and capability gaps in greening supply chains. Conversely, “corruption and external pressure” (SC6) and “pressure to reduce product prices” (EF7) receive the lowest scores, indicating minimal prioritization in the collective decision context. These outcomes reinforce the need for strategic attention to internal competencies and technological infrastructure as they appear to pose the most pressing obstacles from both evaluative perspectives.

To complement the numerical results presented in *Table 9* visually, *Fig. 4* illustrates the final prioritization of GSCM barriers based on Copeland aggregation. This bar chart offers a clear comparative view of each barrier’s relative importance by mapping its Copeland score, thereby facilitating an intuitive understanding of their dominance relationships across combined EDAS and COPRAS evaluations.

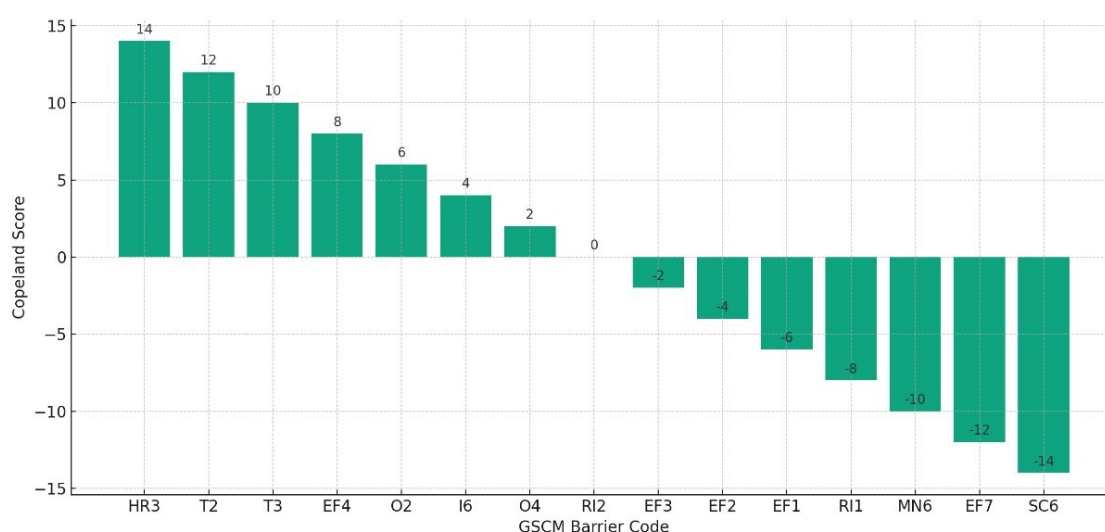


Fig 4. Final barrier prioritization based on Copeland aggregation.

This bar chart presents the final priority ranking of all 15 GSCM barriers using the Copeland aggregation. Each bar represents the Copeland score of a barrier, which

reflects the net number of wins in pairwise comparisons with other barriers across both the EDAS and COPRAS methods. Following the chart, it becomes evident that the barriers “HR3: Inadequate education on sustainable practices,” “T2: Lack of advanced sustainable technologies,” and “T3: Insufficient R&D investment” dominate the prioritization spectrum, achieving the highest Copeland scores. These barriers consistently outperform others in both MADM frameworks, signaling a strategic need to invest in knowledge, innovation, and technological upgrading. In contrast, barriers such as “SC6: Corruption and external pressure” and “EF7: Pressure to reduce product prices” appear at the bottom of the ranking, indicating lower concern from a consensus-based perspective. This visual depiction strengthens the argument for focusing policy and managerial interventions on internal, capability-driven challenges, rather than external economic pressures alone.

4.4|Managerial and Theoretical Implications

The integrated fuzzy multimethod analysis conducted in this study provides practical insights and theoretical contributions to the field of sustainable supply chain management. From a managerial perspective, the results highlight the critical importance of internal capabilities, particularly those related to education, technology, and innovation, as key enablers of successful GSCM implementation. The final prioritization based on Copeland aggregation revealed that the most influential barriers are: I. HR3: Inadequate education on sustainable practices.

II. T2: Lack of advanced sustainable technologies.

III. T3: Insufficient R&D investment.

These consistently top-ranked factors suggest that firms must prioritize long-term capacity building over short-term cost minimization. Investing in sustainability-focused training, accelerating R&D aligned with green objectives, and upgrading technological infrastructure should be considered foundational prerequisites for achieving broader supply chain sustainability. These findings serve as a data-driven roadmap for strategic interventions by industry practitioners, particularly within the manufacturing and logistics sectors. By addressing knowledge gaps and innovation deficiencies, firms are more likely to observe cascading positive effects across the operational, environmental, and customer-facing domains. The relatively lower priority assigned to barriers such as EF7 (Pressure to reduce product prices) and SC6 (Corruption and external pressure) further reinforces the idea that root causes lie within internal capabilities and not external market volatility.

From a theoretical standpoint, this study contributes to the literature by demonstrating the value of hybrid fuzzy MADM approaches for complex sustainability decision problems. The joint application of Fuzzy EDAS and Fuzzy COPRAS, followed by Copeland-based consensus ranking, provides a more balanced and resilient prioritization framework. The complementary nature of distance-based and proportional reasoning enables the extraction of robust priority patterns despite the uncertainty in expert evaluations. Furthermore, the integration of causal-structural inputs (Via fuzzy DEMATEL) into the prioritization process elevates the theoretical rigor of the model, helping differentiate systemic drivers from symptomatic consequences. This underscores a broader insight: sustainable decision making in emerging economies is not merely an economic or environmental optimization problem; it is fundamentally a question of institutional

maturity, learning culture, and technological adaptability. These insights provide a solid foundation for future research on behavioral sustainability modeling, strategic analytics, and capability-driven green transformation, particularly in developing market contexts where structural constraints are intricately linked to operational decisions.

5|Conclusion

This study presents a comprehensive MADM framework to evaluate and prioritize the key barriers to GSCM implementation in the manufacturing sector. By integrating fuzzy logic with two complementary MADM techniques, Fuzzy EDAS and Fuzzy COPRAS, and aggregating the results through the Copeland method, we provided a robust, consensus-based ranking of the 15 identified barriers. The findings highlight that humancentric and knowledge-based challenges, particularly inadequate education on sustainable practices (HR3), insufficient R&D investment (T2, T3), and lack of top management commitment (O2), are the most critical barriers. These consistently ranked highest across all methods, signifying the urgency for strategic interventions in organizational culture, innovation capability, and leadership engagement. Conversely, barriers such as corruption and external pressure (SC6) and pressure to reduce product prices (EF7) were less influential, suggesting that their role may be more symptomatic than causal. From a methodological standpoint, this study demonstrates the value of hybrid fuzzy MADM models for sustainability decision problems, particularly when data are ambiguous or derived from linguistic expert input. The Copeland aggregation strategy adds a layer of rigor by reconciling divergent rankings from the EDAS and COPRAS, thereby enhancing the reliability of decisions. Several avenues for future research exist. First, the integration of other MADM techniques, such as fuzzy MARCOS, VIKOR, and BWM, can be explored to test consistency and robustness further. Second, longitudinal studies can investigate whether the criticality of barriers changes over time as GSCM practices mature. Third, scenario-based extensions using fuzzy cognitive mapping or system dynamics can reveal interdependencies between barriers in dynamic environments. Finally, expanding the framework to cross-sectoral or multinational comparisons would help generalize the findings and adapt policy recommendations to various industrial and cultural contexts. By offering a structured and data-driven evaluation of GSCM barriers, this study contributes to both theoretical advancement and actionable managerial insight in the field of sustainable operations.

Author Contribution

Conceptualization, T. C. and N.A.J.; Methodology, T.C., Software, Y. B. Validation: W.L., Y.B., and H.M.; Formal analysis, T.C.; Investigation, T.C. Resources, N. A. J. Data Maintenance, W. L. Writing —original draft preparation, T. C. Writing —review and editing: N.A.J. and W.L. Visualization: H. M. Supervision, N. A. J. Project administration, T. C. Funding acquisition: N.A.J. All the authors have read and agreed to the published version of the manuscript. Authorship is restricted to individuals who make significant contributions to the work. All roles were defined according to the Contributor Roles Taxonomy (CRediT) file.

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Data Availability

All data generated or analyzed in this study are included in this published article. For further inquiries, please contact the corresponding authors.

Conflicts of Interest

The authors declare that they have no conflicts of interest relevant to the content of this article.

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